

FTML practical session 11

11 juin 2026

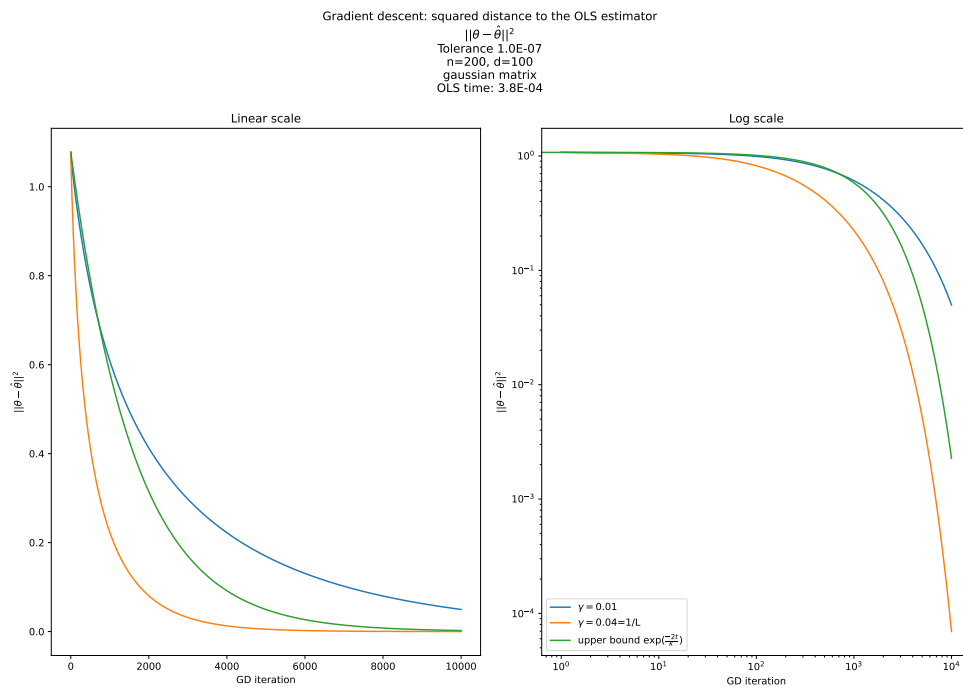


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1 CONVERGENCE SPEED OF GRADIENT DESCENT

1.1 Setting

We want to study the speed of convergence of the minimization of a convex function f defined over $\mathbb{R}^d$, with gradient descent.

$$\forall t, \theta_{t+1} \leftarrow \theta_t - \gamma \nabla_{\theta} f(\theta_t) \quad (1)$$

where t is the iteration index.

We will study the specific case of linear regression (OLS), but the results partially generalize to general convex functions. We use the usual objects :

- design matrix $X \in \mathbb{R}^{n,d}$
- label vector $y \in \mathbb{R}^n$.
- loss function

$$f(\theta) = \frac{1}{2n} \|X\theta - y\|^2 \quad (2)$$

- $\|\cdot\|$ is the usual euclidean norm.

The gradient and the Hessian write :

$$\nabla_{\theta} f(\theta) = \frac{1}{n} X^T (X\theta - y) \quad (3)$$

$$H = \frac{1}{n} X^T X \quad (4)$$

We note θ^* the minimizers of f . All minimizers verify that

$$\nabla_{\theta} f(\theta^*) = 0 \quad (5)$$

or

$$H\theta^* = \frac{1}{n} X^T y \quad (6)$$

If H is not invertible, they might be not unique, but all have the same function value $f(\theta^*)$.

- H is symmetric, positive semi-definite.
- H is invertible if and only if its smallest eigenvalue μ is > 0 , in which case f is strongly convex (see section 2.3 in https://github.com/nlehir/FTML/blob/master/lecture_notes/lecture%20notes.pdf)

1.2 Convergence speed of gradient descent

We assume that $\mu > 0$, meaning that H is invertible. Let us study the convergence speed of GD towards θ^* (that exists and is unique).

1.2.1 Step 1

Show that

$$\forall \theta \in \mathbb{R}^d, f(\theta) - f(\theta^*) = \frac{1}{2} (\theta - \theta^*)^T H (\theta - \theta^*) \quad (7)$$

1.2.2 Step 2

Show that

$$\forall t \in \mathbb{N}, \theta_t = \theta_{t-1} - \gamma H (\theta_{t-1} - \theta^*) \quad (8)$$

1.2.3 Step 3

Deduce that :

$$\theta_t - \theta^* = (I - \gamma H)(\theta_{t-1} - \theta^*) \quad (9)$$

and that

$$\theta_t - \theta^* = (I - \gamma H)^t(\theta_0 - \theta^*) \quad (10)$$

where θ_0 is the initial value of θ .

1.2.4 Step 4

We can use two measures of performance of the gradient algorithm. Using the previous results, they write :

— Distance to minimizer :

$$\|\theta_t - \theta^*\|^2 = (\theta_0 - \theta^*)^T (I - \gamma H)^{2t} (\theta_0 - \theta^*) \quad (11)$$

— Convergence in function values :

$$f(\theta_t) - f(\theta^*) = \frac{1}{2} (\theta_0 - \theta^*)^T (I - \gamma H)^{2t} H (\theta_0 - \theta^*) \quad (12)$$

We introduce the **condition number** $\kappa = \frac{L}{\mu}$ where L is the largest eigenvalue of H . By convention, if $\mu = 0$, $L = +\infty$. Show that with a good choice of γ , we obtain an **exponential convergence**

$$\|\theta_t - \theta^*\|^2 \leq \left(1 - \frac{1}{\kappa}\right)^{2t} \|\theta_0 - \theta^*\|^2 \quad (13)$$

We note that

$$\left(1 - \frac{1}{\kappa}\right)^{2t} \leq \exp\left(-\frac{1}{\kappa}\right)^{2t} = \exp\left(-\frac{2t}{\kappa}\right) \quad (14)$$

1.2.5 Simulation

Run a simulation that plots both the upper bound and the convergence speed of GD on a least squares problem, like seen on Figure 1. You can adapt the code from `tp_o6_line_search`.

1.2.6 Non strongly convex functions

If $\mu = 0$, we do not have an exponential convergence guarantee, but rather a convergence rate in $\mathcal{O}\left(\frac{1}{t}\right)$ (the proof is different).

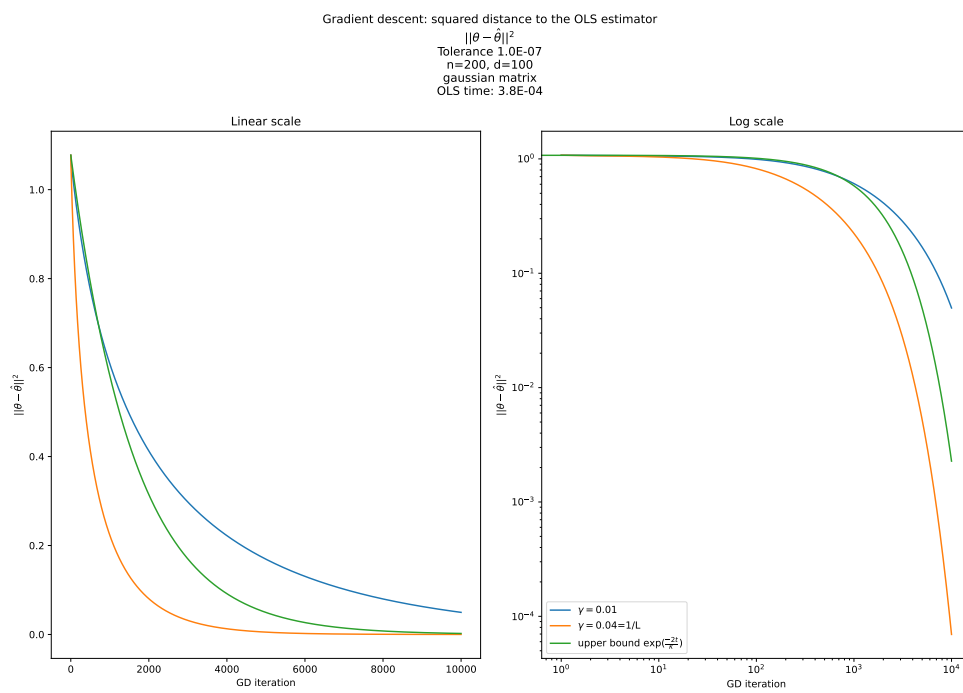


FIGURE 1 – Comparison of the upper bound and of the actual convergence speed.